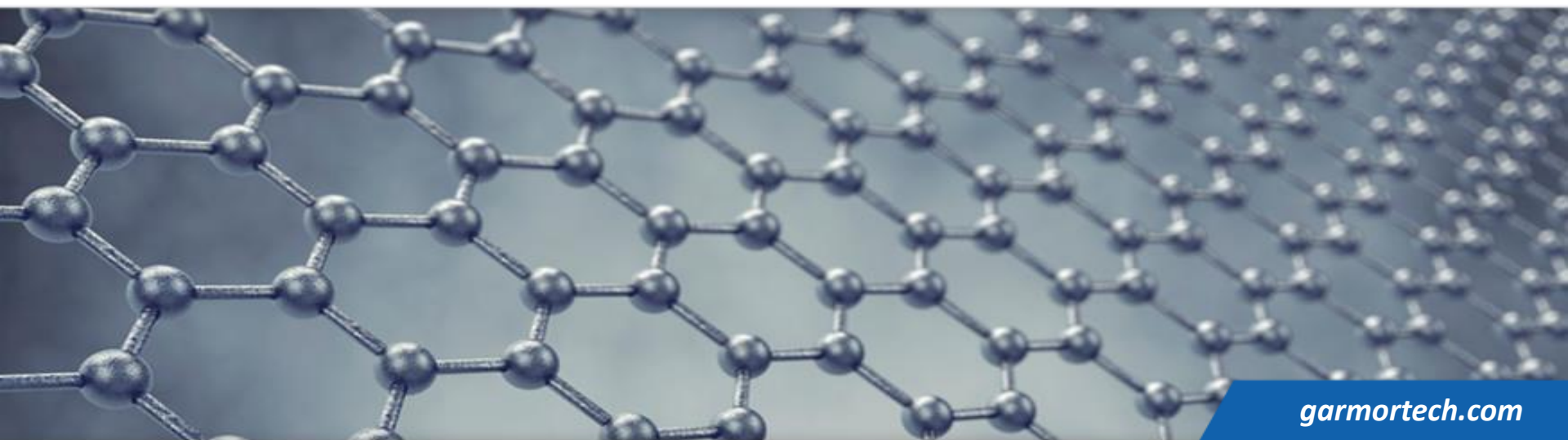




INNOVATIVE SOLUTIONS. *ADVANCED PRODUCTS.*



garmortech.com

Delivering Affordable Few-Layer Graphene Oxide for High-Volume Applications

Sean Christiansen, PhD, Vice President of Engineering
November 18, 2015

More Than Just a Graphene Company



Manufacturing

- High-quality, low-cost graphene oxide
- Expertise in handling/dispersing GO for composite materials



Plug and Play Systems

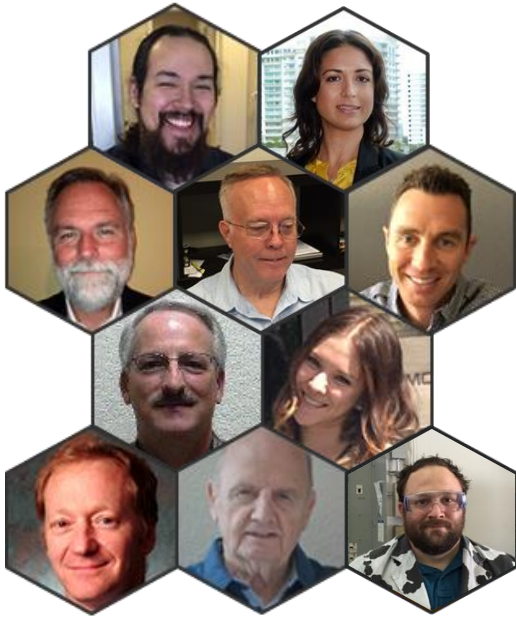
- “Green” production process
- Turn-key systems supported by embedded controls
- For high-volume, price-sensitive applications



Material Development

- Novel materials created using GO and rGO
- Customer driven

The Team



Executive Team

- Experienced commercializing university IP
- Raised \$100MM+ in early-stage funding for prior firms
- Multiple successful exits including pending IPO

Business Development Team

- Former DARPA experience transitioning technology to commercial status
- Multiple personnel with 30+ years average experience generating sales at Fortune 500 companies

Technical Team

- Four PhD chemists and chemical engineers
- Specialist in dispersion and blending technologies
- Six patents pending; ten provisional patents filed

Graphene Applications

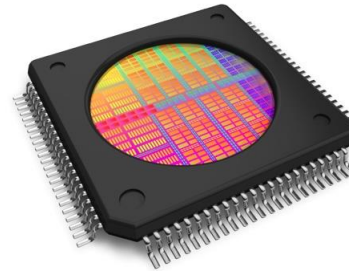
Single Layer Graphene (Future Products)



Cell Phones



Flexible Displays



Integrated Circuits



Flake Graphene Oxide (GO) (address today's markets)



Coatings



Plastics Additive



Cement Admixture



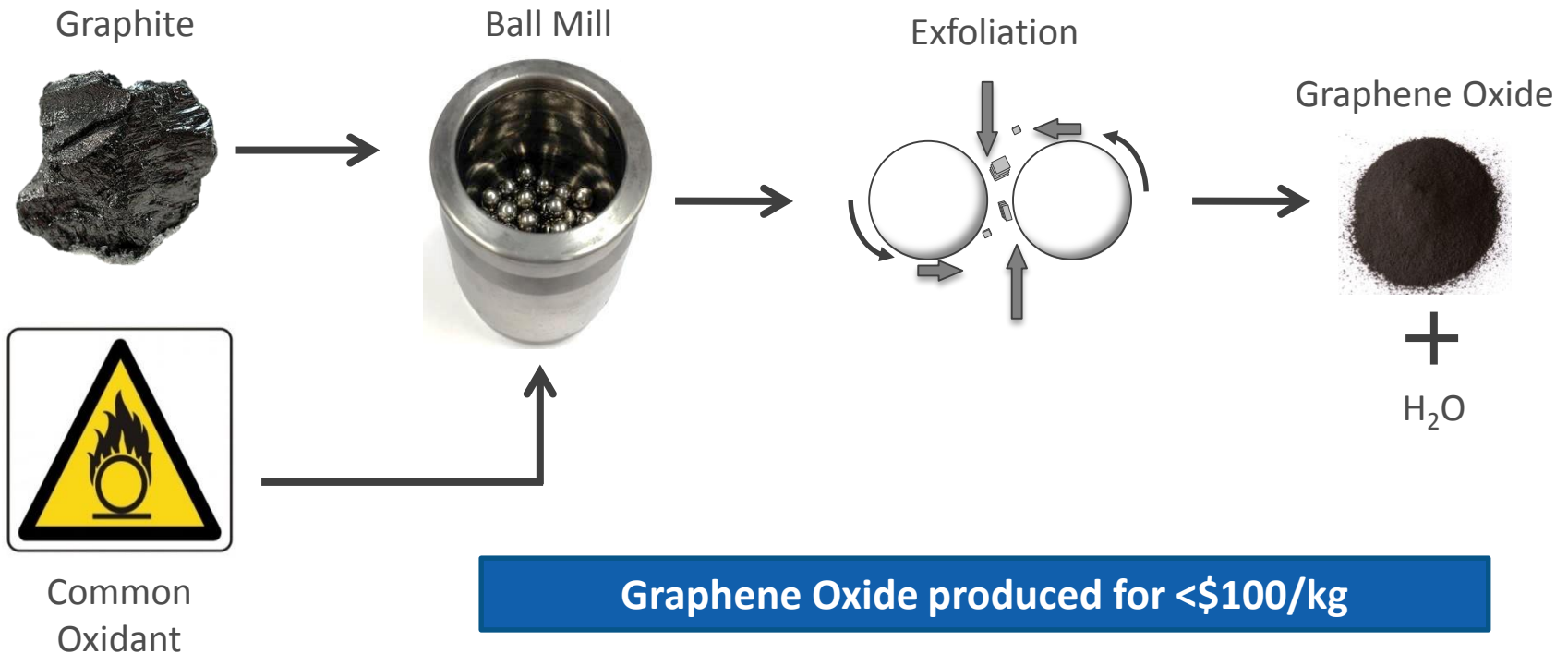
Thin Films

Bipolar Plates

The Garmor Solution

Graphene oxide created using an inexpensive, green process

- Manufacturing tools have been in use for over 100 years → **LOW STARTUP COST**
- Technique uses common reactants and produces no waste → **MINIMAL OPERATING COST**
- Process can be located on-site → **NO TRANSPORTATION COST**



GO Production

10,000 ft² flex space

- Multiple bay doors facilitate logistics

Low-rate production machine

- 1kg graphene oxide/day
- Highly instrumented machine

High-rate production machine

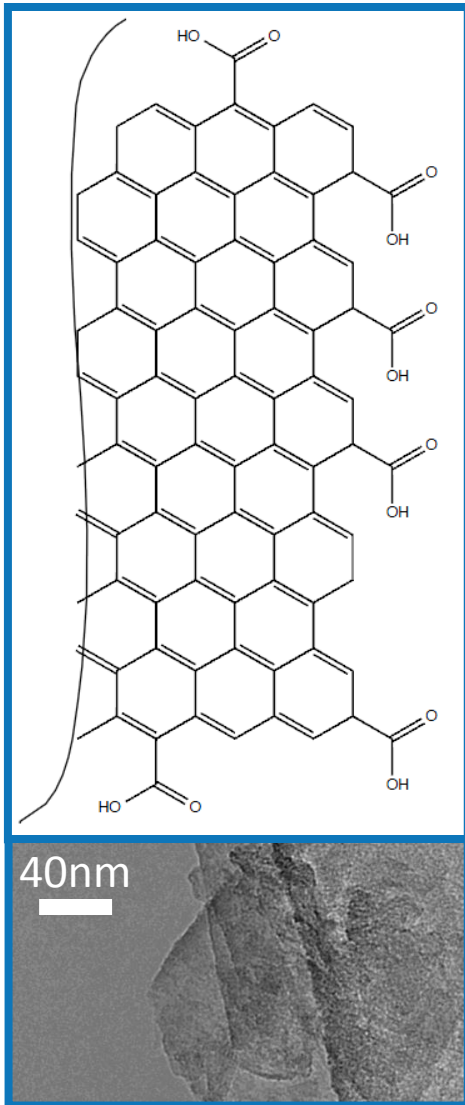
- 100 metric tons graphene oxide/year
- Customer validation of process

Quality Control

- Oxygen analysis
- Thermogravimetric analysis (TGA)
- X-ray diffraction
- Dynamic Light Scattering
- TEM / SEM
- AFM



Two “solutions” for any “solution”

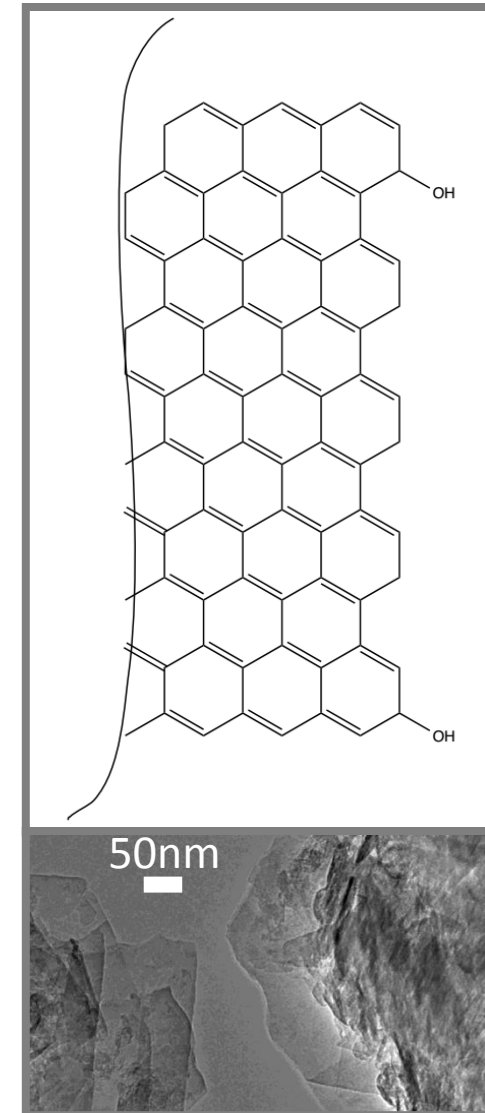


← Graphene Oxide

- “As made” material without any post-treatment
- Subjected to drying process
- 5-15% Oxygen
- Polar/hydrophilic
- Ideal for polar hosts and cement

Reduced GO →

- Graphene oxide subjected to a reduction step to remove oxygen
- 1-3% Oxygen
- Non-polar/Hydrophobic
- Ideal for non-polar hosts
- Ideal for corrosion inhibition



Downstream Development Efforts

GO-Polymers: Electrical Conductivity

Graphene offers ballistic conduction (6,000 S/cm)

- Conductivity can be tailored by loading percentage
- Low loadings facilitate processability

Conductive graphene coatings for FRP tanks, pipelines, inks, and parts

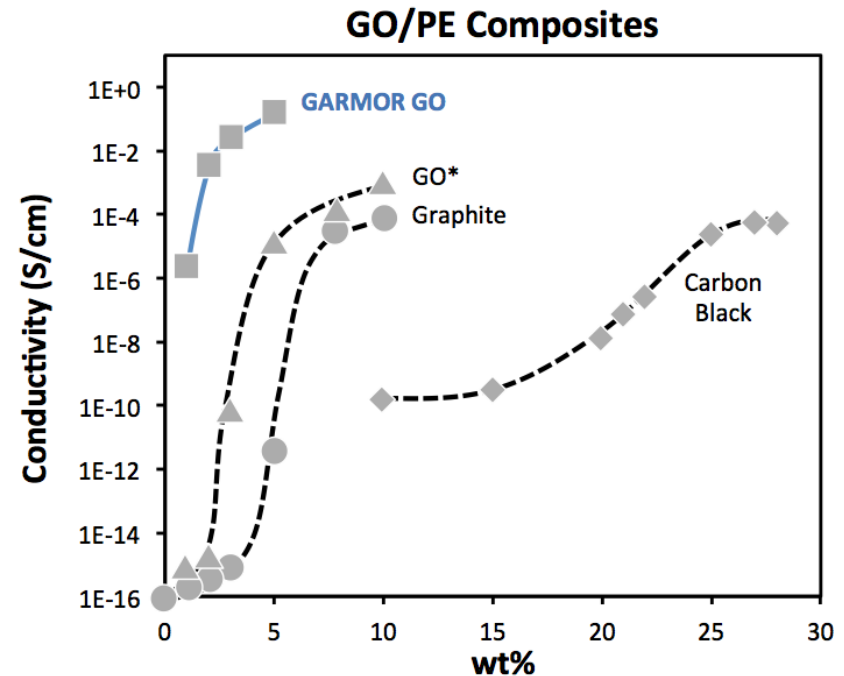
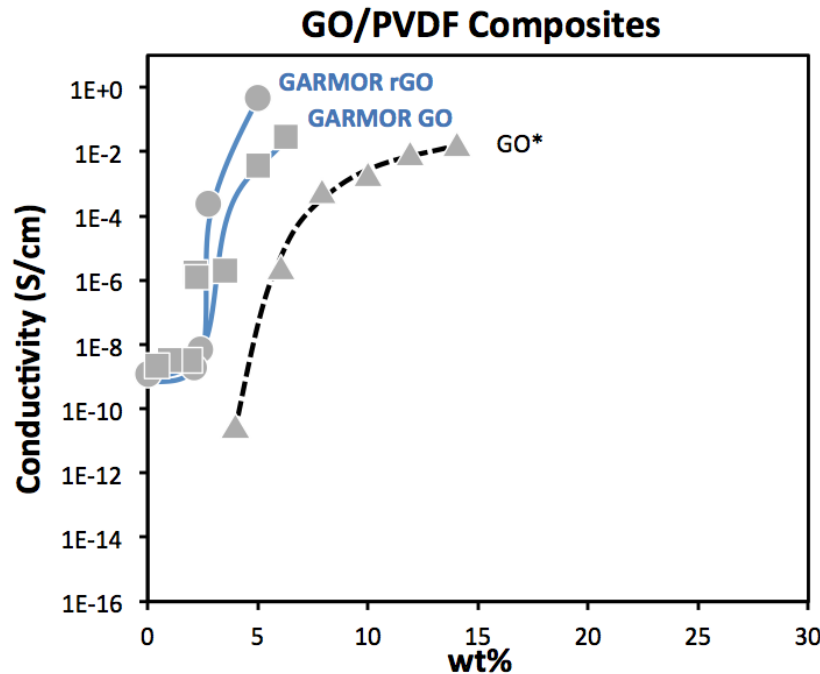


Electrostatic Control



EMI Shielding

GO-Polymers: Electrical Conductivity



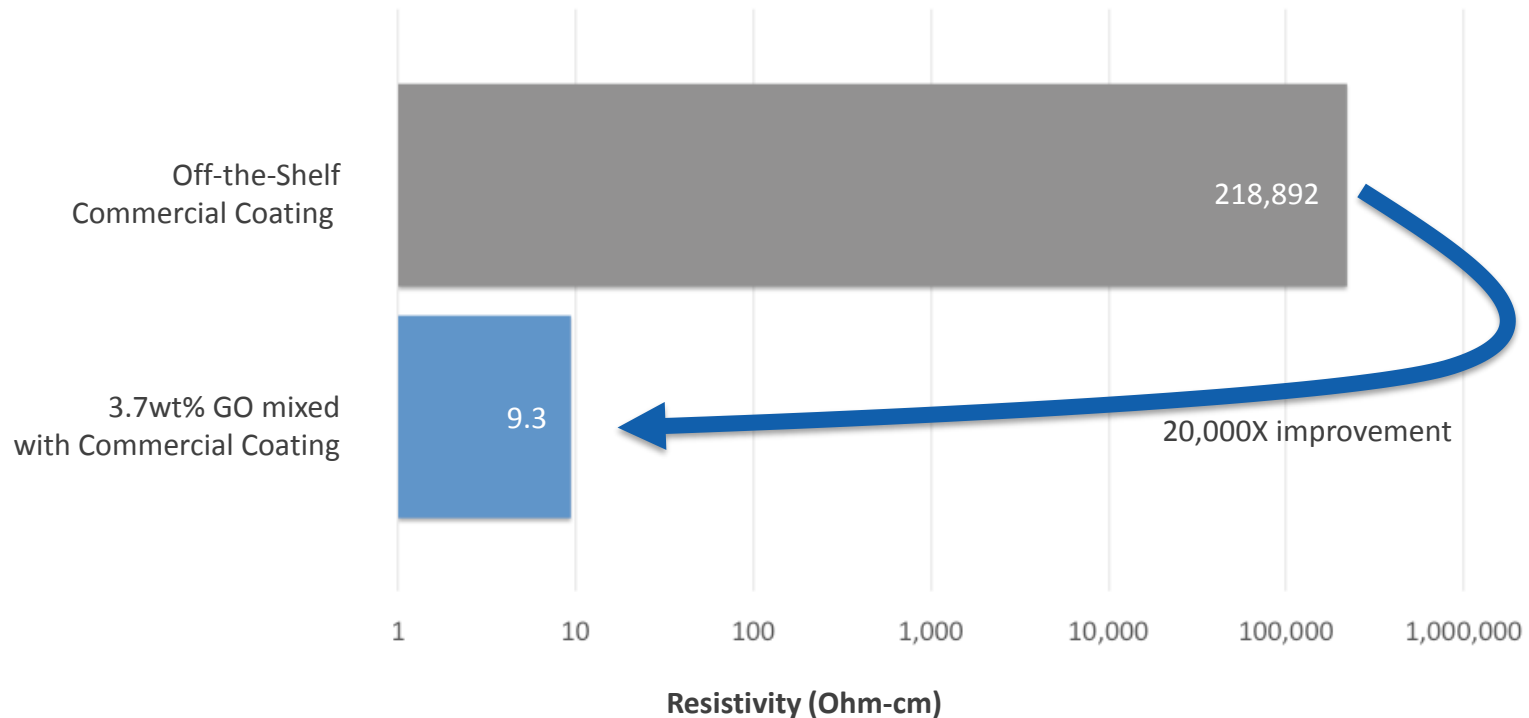
Bulk electrical conductivity of the composites were measured using an electrometer workstation.

Values on the graphs represent an average of three values recorded for each GO loading.

GO*: Graphene oxide prepared via sonication of graphite.

Garmor GO demonstrates significant performance advantage

GO-Polymers: ESD Applications



Bulk electrical conductivity values of the materials represent an average of three values recorded on distinct samples. The GO-coating sample was subjected to a Garmor mixing method for uniformly dispersing the GO throughout the material.

Improved dispersion methods aid coating development

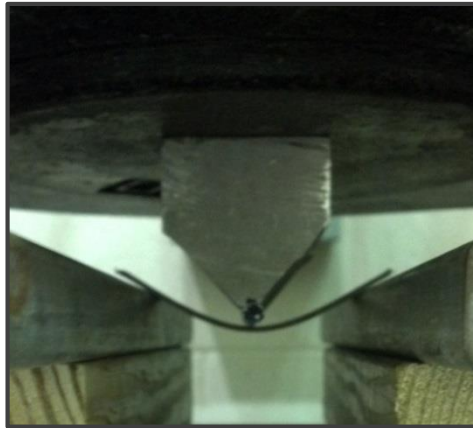
GO-Fiberglass: Fatigue Life

Graphene: Inhibits direct crack propagation

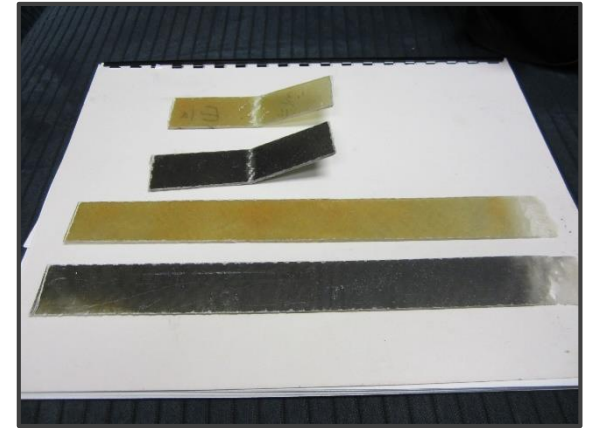
- 0.02wt% GO sprayed on fiberglass laminate
 - Substantial increase in uniaxial flexural fatigue life
 - Increase in tension fatigue life
 - Increase in flexural strength
- Strong interfacial bond to epoxy
- Simple process could be easily adapted to existing layups



Fiberglass Forming in Vacuum Bag



Flexural Testing

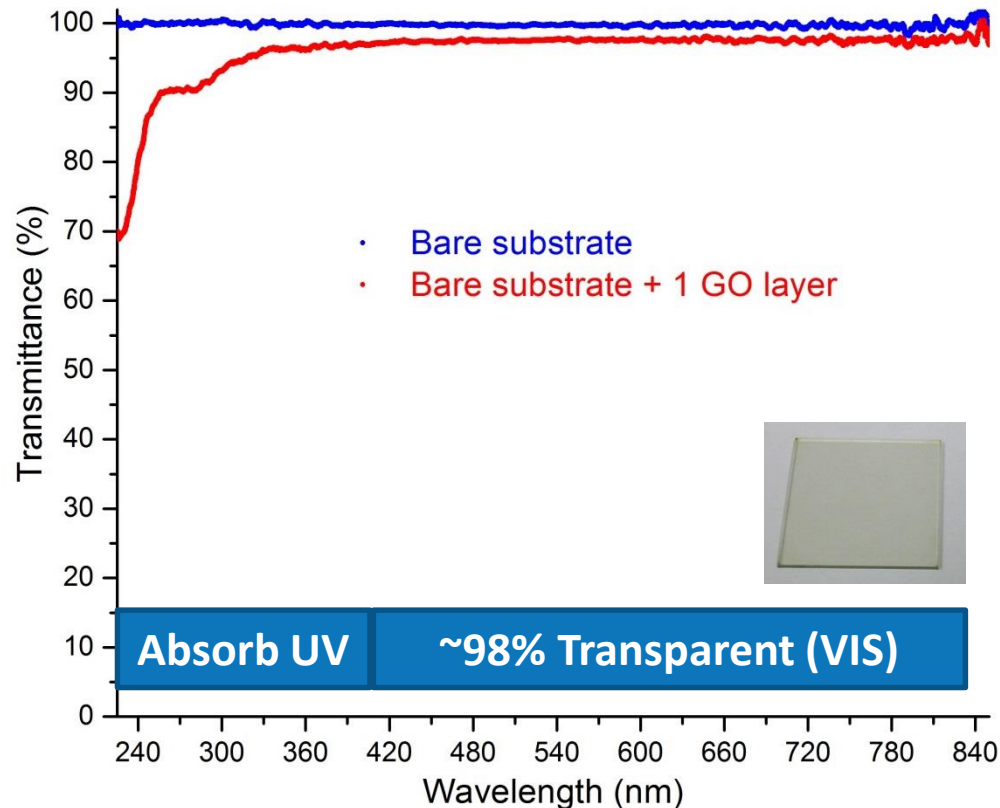


Neat Fiberglass and GO Fiberglass

Thin GO Films

Graphene oxide films for UV protection

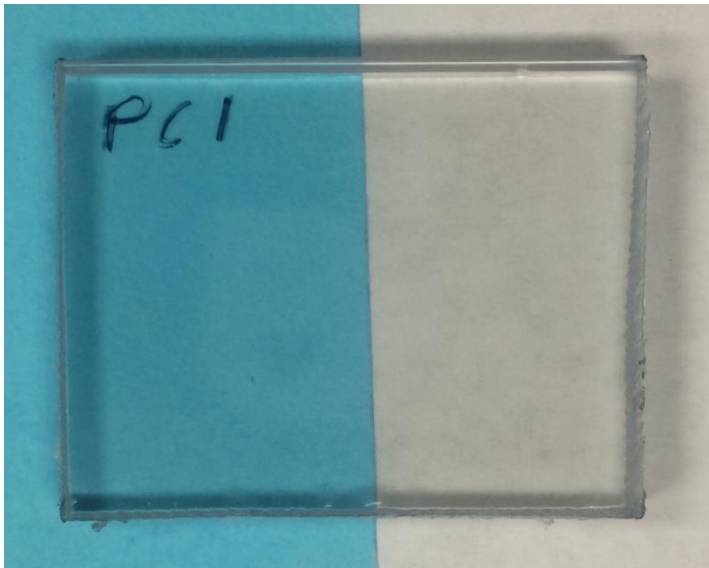
- GO: transparent in the visible spectrum yet absorbs in the UV range
- Opportunities in automotive and construction for UV protection



Thin GO Films

Program Objectives

- Coating that prevents UV-induced yellowing in polycarbonate
- The potential for an electrically resistive heating element to act as a lens defogger/defrosting element
- Provide a cost-effective, high volume manufacturing by dispersion of The GO in the PC lens protective hard coat resin



Four layers of graphene oxide coatings on PC

Methods

- Spray, spin and dip coating of GO and GO mixtures
- Evaluated on various host materials

New Material Development

GO-PAN

Advanced composite material created by Garmor

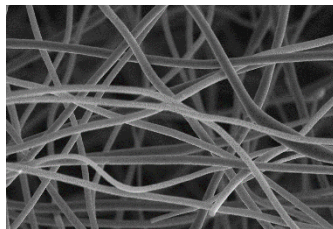
- Leverages precursor for carbon fiber in small amounts
- Conductive, strong, ultra-lightweight composite
- Easier to shape and extrude/stamped into a final form

Graphene Oxide



90%

Polyacrylonitrile



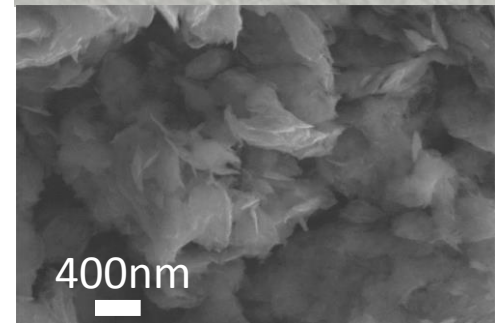
10wt%

+

Heat

Pressure

GO-PAN



400nm

$\sigma > 100 \text{ S/cm}$

**Designed for mass-production extrusion
and stamping methods**

THANK YOU

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GARMOR